

B5 General Relativity — Model Solutions, Michaelmas Term 2020

Question 1 — Affine connection, covariant derivative, Riemann tensor

Problem. (a) Define $\Gamma_{\mu\nu}^\lambda$ via locally free-falling coordinates and write $V_{;\nu}^\mu$. (b) Show $\Gamma_{\mu\nu}^\lambda$ is not a tensor. (c) Show $V_{;\sigma;\tau}^\mu - V_{;\tau;\sigma}^\mu = R_{\lambda\sigma\tau}^\mu V^\lambda$. (d) Argue $R_{\lambda\sigma\tau}^\mu$ is therefore a tensor. (e) Relations between $R_{\lambda\sigma\tau}^\mu$, $R_{\lambda\mu\sigma\tau}$, $R_{\mu\nu}$, R ; symmetries of $R_{\lambda\mu\sigma\tau}$.

(a) Affine connection from local inertial frame

Erect locally free-falling (LIF) coordinates ξ^α at event P . Free particles obey:

$$\frac{d^2\xi^\alpha}{d\tau^2} = 0.$$

Apply the chain rule to $\xi^\alpha(x^\mu)$:

$$\frac{d\xi^\alpha}{d\tau} = \frac{\partial\xi^\alpha}{\partial x^\mu} \frac{dx^\mu}{d\tau}.$$

Differentiate once more using the product rule:

$$\frac{d^2\xi^\alpha}{d\tau^2} = \frac{\partial\xi^\alpha}{\partial x^\mu} \frac{d^2x^\mu}{d\tau^2} + \frac{\partial^2\xi^\alpha}{\partial x^\mu \partial x^\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}.$$

Set the left side to zero:

$$\frac{\partial\xi^\alpha}{\partial x^\mu} \frac{d^2x^\mu}{d\tau^2} + \frac{\partial^2\xi^\alpha}{\partial x^\mu \partial x^\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.$$

Multiply through by $\partial x^\lambda / \partial \xi^\alpha$ and use $(\partial x^\lambda / \partial \xi^\alpha)(\partial \xi^\alpha / \partial x^\mu) = \delta_\mu^\lambda$:

$$\frac{d^2x^\lambda}{d\tau^2} + \frac{\partial x^\lambda}{\partial \xi^\alpha} \frac{\partial^2\xi^\alpha}{\partial x^\mu \partial x^\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0.$$

Define the affine connection:

$$\Gamma_{\mu\nu}^\lambda \equiv \frac{\partial x^\lambda}{\partial \xi^\alpha} \frac{\partial^2\xi^\alpha}{\partial x^\mu \partial x^\nu}. \quad (1)$$

Geodesic equation becomes $\ddot{x}^\lambda + \Gamma_{\mu\nu}^\lambda \dot{x}^\mu \dot{x}^\nu = 0$.

Covariant derivative of a vector V^μ :

$$V_{;\nu}^\mu = \frac{\partial V^\mu}{\partial x^\nu} + \Gamma_{\lambda\nu}^\mu V^\lambda. \quad (2)$$

The added Γ -term cancels the non-tensorial piece of $\partial_\nu V^\mu$ so that $V_{;\nu}^\mu$ transforms as a $(1,1)$ tensor.

(b) $\Gamma_{\mu\nu}^\lambda$ is not a tensor

Keep the LIF ξ^α but change chart $x^\mu \rightarrow x'^\mu$. Apply definition (1) in primed coordinates:

$$\Gamma_{\mu\nu}^{\prime\lambda} = \frac{\partial x'^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x'^\mu \partial x'^\nu}.$$

Expand the first derivative by chain rule:

$$\frac{\partial \xi^\alpha}{\partial x'^\nu} = \frac{\partial x^\rho}{\partial x'^\nu} \frac{\partial \xi^\alpha}{\partial x^\rho}.$$

Differentiate w.r.t. x'^μ using the product rule:

$$\frac{\partial^2 \xi^\alpha}{\partial x'^\mu \partial x'^\nu} = \frac{\partial x^\sigma}{\partial x'^\mu} \frac{\partial x^\rho}{\partial x'^\nu} \frac{\partial^2 \xi^\alpha}{\partial x^\sigma \partial x^\rho} + \frac{\partial^2 x^\rho}{\partial x'^\mu \partial x'^\nu} \frac{\partial \xi^\alpha}{\partial x^\rho}.$$

Contract with $\partial x'^\lambda / \partial \xi^\alpha$ and apply the chain rule on the inhomogeneous piece:

$$\frac{\partial x'^\lambda}{\partial \xi^\alpha} \frac{\partial \xi^\alpha}{\partial x^\rho} = \frac{\partial x'^\lambda}{\partial x^\rho}.$$

Insert an intermediate $\partial x^\kappa / \partial \xi^\alpha \cdot \partial x'^\lambda / \partial x^\kappa$ in the homogeneous piece to recognise $\Gamma_{\sigma\rho}^\kappa$:

$$\frac{\partial x'^\lambda}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x^\sigma \partial x^\rho} = \frac{\partial x'^\lambda}{\partial x^\kappa} \frac{\partial x^\kappa}{\partial \xi^\alpha} \frac{\partial^2 \xi^\alpha}{\partial x^\sigma \partial x^\rho} = \frac{\partial x'^\lambda}{\partial x^\kappa} \Gamma_{\sigma\rho}^\kappa.$$

Combine:

$$\boxed{\Gamma_{\mu\nu}^{\prime\lambda} = \frac{\partial x'^\lambda}{\partial x^\kappa} \frac{\partial x^\sigma}{\partial x'^\mu} \frac{\partial x^\rho}{\partial x'^\nu} \Gamma_{\sigma\rho}^\kappa + \frac{\partial x'^\lambda}{\partial x^\rho} \frac{\partial^2 x^\rho}{\partial x'^\mu \partial x'^\nu}.} \quad (3)$$

First term is the tensor transformation; the second (inhomogeneous) piece is non-zero whenever the coordinate transformation is non-linear, so $\Gamma_{\mu\nu}^\lambda$ is **not a tensor**.

(c) Commutator of covariant derivatives

Apply (2) once on V^μ :

$$V_{;\sigma}^\mu = \partial_\sigma V^\mu + \Gamma_{\lambda\sigma}^\mu V^\lambda.$$

Treat $V_{;\sigma}^\mu$ as a (1,1) tensor and apply the rule on covariant derivatives of mixed tensors:

$$V_{;\sigma;\tau}^\mu = \partial_\tau V_{;\sigma}^\mu + \Gamma_{\nu\tau}^\mu V_{;\sigma}^\nu - \Gamma_{\sigma\tau}^\nu V_{;\nu}^\mu.$$

Substitute $V_{;\sigma}^\mu = \partial_\sigma V^\mu + \Gamma_{\lambda\sigma}^\mu V^\lambda$:

$$V_{;\sigma;\tau}^\mu = \partial_\tau (\partial_\sigma V^\mu + \Gamma_{\lambda\sigma}^\mu V^\lambda) + \Gamma_{\nu\tau}^\mu (\partial_\sigma V^\nu + \Gamma_{\lambda\sigma}^\nu V^\lambda) - \Gamma_{\sigma\tau}^\nu (\partial_\nu V^\mu + \Gamma_{\lambda\nu}^\mu V^\lambda).$$

Expand the first parenthesis via the product rule:

$$V_{;\sigma;\tau}^\mu = \partial_\tau \partial_\sigma V^\mu + (\partial_\tau \Gamma_{\lambda\sigma}^\mu) V^\lambda + \Gamma_{\lambda\sigma}^\mu \partial_\tau V^\lambda + \Gamma_{\nu\tau}^\mu (\partial_\sigma V^\nu + \Gamma_{\lambda\sigma}^\nu V^\lambda) - \Gamma_{\sigma\tau}^\nu (\partial_\nu V^\mu + \Gamma_{\lambda\nu}^\mu V^\lambda).$$

Swap $\sigma \leftrightarrow \tau$ to obtain the partner:

$$V_{;\tau;\sigma}^\mu = \partial_\sigma \partial_\tau V^\mu + (\partial_\sigma \Gamma_{\lambda\tau}^\mu) V^\lambda + \Gamma_{\lambda\tau}^\mu \partial_\sigma V^\lambda + \Gamma_{\nu\sigma}^\mu (\partial_\tau V^\nu + \Gamma_{\lambda\tau}^\nu V^\lambda) - \Gamma_{\tau\sigma}^\nu (\partial_\nu V^\mu + \Gamma_{\lambda\nu}^\mu V^\lambda).$$

Commute the second partials so they cancel in the difference:

$$\partial_\tau \partial_\sigma V^\mu - \partial_\sigma \partial_\tau V^\mu = 0.$$

Apply torsion-free symmetry $\Gamma_{\sigma\tau}^\nu = \Gamma_{\tau\sigma}^\nu$ to kill the last bracket of each line:

$$\Gamma_{\sigma\tau}^\nu (\partial_\nu V^\mu + \Gamma_{\lambda\nu}^\mu V^\lambda) - \Gamma_{\tau\sigma}^\nu (\partial_\nu V^\mu + \Gamma_{\lambda\nu}^\mu V^\lambda) = 0.$$

Collect the $\Gamma \cdot \partial V$ cross-terms from the two expressions:

$$(\Gamma_{\lambda\sigma}^\mu \partial_\tau V^\lambda + \Gamma_{\nu\tau}^\mu \partial_\sigma V^\nu) - (\Gamma_{\lambda\tau}^\mu \partial_\sigma V^\lambda + \Gamma_{\nu\sigma}^\mu \partial_\tau V^\nu).$$

Relabel dummies $\nu \rightarrow \lambda$ in both inner brackets:

$$\Gamma_{\lambda\sigma}^\mu \partial_\tau V^\lambda + \Gamma_{\lambda\tau}^\mu \partial_\sigma V^\lambda - \Gamma_{\lambda\tau}^\mu \partial_\sigma V^\lambda - \Gamma_{\lambda\sigma}^\mu \partial_\tau V^\lambda = 0.$$

Only the $\partial\Gamma$ and $\Gamma\Gamma$ brackets remain:

$$V_{;\sigma;\tau}^\mu - V_{;\tau;\sigma}^\mu = [\partial_\tau \Gamma_{\lambda\sigma}^\mu - \partial_\sigma \Gamma_{\lambda\tau}^\mu + \Gamma_{\nu\tau}^\mu \Gamma_{\lambda\sigma}^\nu - \Gamma_{\nu\sigma}^\mu \Gamma_{\lambda\tau}^\nu] V^\lambda.$$

Identify the bracket with the Riemann tensor (formula on the cover sheet):

$$\boxed{V_{;\sigma;\tau}^\mu - V_{;\tau;\sigma}^\mu = R_{\lambda\sigma\tau}^\mu V^\lambda.} \quad (4)$$

(d) $R_{\lambda\sigma\tau}^\mu$ is a tensor

LHS of (4) is a $(1,3)$ tensor: each covariant derivative produces a tensor, and a difference of tensors is a tensor. RHS is the contraction of $R_{\lambda\sigma\tau}^\mu$ with the arbitrary vector V^λ . By the **quotient theorem**, since $R_{\lambda\sigma\tau}^\mu V^\lambda$ is a tensor for every V^λ , $R_{\lambda\sigma\tau}^\mu$ itself is a $(1,3)$ tensor. The non-tensorial inhomogeneous pieces in the individual $\partial\Gamma$ and $\Gamma\Gamma$ terms cancel in the antisymmetric combination.

(e) Index relations and symmetries

Lower the upper index with $g_{\mu\lambda}$:

$$R_{\mu\lambda\sigma\tau} = g_{\mu\nu} R_{\lambda\sigma\tau}^\nu.$$

Ricci tensor (contract first upper with last lower, per cover-sheet convention):

$$R_{\mu\nu} \equiv R_{\mu\nu\lambda}^\lambda = g^{\lambda\sigma} R_{\sigma\mu\nu\lambda}.$$

Ricci scalar:

$$R \equiv g^{\mu\nu} R_{\mu\nu}.$$

Symmetries of $R_{\lambda\mu\sigma\tau}$:

$$\boxed{R_{\lambda\mu\sigma\tau} = -R_{\mu\lambda\sigma\tau} = -R_{\lambda\mu\tau\sigma} = R_{\sigma\tau\lambda\mu}, \quad R_{\lambda[\mu\sigma\tau]} = 0, \quad R_{\lambda\mu[\sigma\tau;\rho]} = 0.}$$

The first three are the algebraic (anti)symmetries; the cyclic identity (first Bianchi) and the differential (second Bianchi) identities complete the set, reducing the $4^4 = 256$ components to 20 in 4D.

Question 2 — Rindler frame, Newtonian limit, time dilation, proper acceleration

Problem. (a) Convert Minkowski to Rindler line element. (b) Newtonian limit. (c) Proper-time ratio at $z' = 0$ vs $z' = h$ giving $1 + gh/c^2$. (d) Equivalence with gravitational time dilation. (e) Compute invariant 4-acceleration a and show it differs at $z' = 0$ and $z' = h$.

(a) Rindler line element

Differentiate ct in the given transformation:

$$c dt = \left(\frac{c^2}{g} + z' \right) \frac{g}{c} \cosh(gt'/c) dt' + \sinh(gt'/c) dz'.$$

Factor c^2/g out of the bracket:

$$c dt = \left(1 + \frac{gz'}{c^2} \right) c \cosh(gt'/c) dt' + \sinh(gt'/c) dz'.$$

Differentiate z analogously:

$$dz = \left(1 + \frac{gz'}{c^2} \right) c \sinh(gt'/c) dt' + \cosh(gt'/c) dz'.$$

Square $c dt$ and dz separately:

$$c^2 dt^2 = \left(1 + \frac{gz'}{c^2} \right)^2 c^2 \cosh^2 dt'^2 + 2 \left(1 + \frac{gz'}{c^2} \right) c \cosh \sinh dt' dz' + \sinh^2 dz'^2,$$

$$dz^2 = \left(1 + \frac{gz'}{c^2} \right)^2 c^2 \sinh^2 dt'^2 + 2 \left(1 + \frac{gz'}{c^2} \right) c \sinh \cosh dt' dz' + \cosh^2 dz'^2.$$

Subtract; the cross-terms cancel and combine the dt'^2, dz'^2 coefficients:

$$-c^2 dt^2 + dz^2 = - \left(1 + \frac{gz'}{c^2} \right)^2 c^2 (\cosh^2 - \sinh^2) dt'^2 + (\cosh^2 - \sinh^2) dz'^2.$$

Apply $\cosh^2 - \sinh^2 = 1$:

$$-c^2 dt^2 + dz^2 = - \left(1 + \frac{gz'}{c^2} \right)^2 c^2 dt'^2 + dz'^2.$$

Add the unchanged $dx^2 + dy^2 = dx'^2 + dy'^2$:

$$\boxed{ds^2 = - \left(1 + \frac{gz'}{c^2} \right)^2 c^2 dt'^2 + dx'^2 + dy'^2 + dz'^2.} \quad (5)$$

(b) Newtonian uniformly accelerated field

Newtonian limit: weak field and slow motion. Require $gz'/c^2 \ll 1$ (and the test motion non-relativistic). Expand g_{00} :

$$g_{00} = - \left(1 + \frac{gz'}{c^2} \right)^2 \approx -1 - \frac{2gz'}{c^2}.$$

Compare with the weak-field metric $g_{00} = -(1 + 2\Phi/c^2)$:

$$\Phi(z') = gz'.$$

Newtonian field:

$$\boxed{\mathbf{a} = -\nabla\Phi = -g \hat{\mathbf{z}}'}$$

A constant uniform acceleration in $-\hat{z}'$, identifiable with a uniformly accelerated reference frame. The condition is $|gz'/c^2| \ll 1$ over the region considered.

(c) Proper-time ratio

Clock at rest at $z' = z'_0$ means $dx' = dy' = dz' = 0$. From (5) the proper-time element is $d\tau^2 = -ds^2/c^2$:

$$d\tau^2 = \left(1 + \frac{gz'_0}{c^2}\right)^2 dt'^2.$$

Take the positive square root:

$$d\tau = \left(1 + \frac{gz'_0}{c^2}\right) dt'.$$

At $z' = 0$:

$$d\tau_0 = dt'.$$

At $z' = h$:

$$d\tau_h = \left(1 + \frac{gh}{c^2}\right) dt'.$$

Take the ratio:

$$\boxed{\frac{d\tau_h}{d\tau_0} = 1 + \frac{gh}{c^2}}. \quad (6)$$

A clock at higher z' ticks faster: it accumulates more proper time per unit coordinate time t' .

(d) Gravitational time dilation

In the weak-field static metric write $g_{tt} = -(1 + 2\Phi/c^2)c^2$. For a clock at rest $dx^i = 0$, so $-c^2 d\tau^2 = g_{tt} dt^2$:

$$d\tau^2 = -\frac{g_{tt}}{c^2} dt^2.$$

Substitute the weak-field form:

$$d\tau^2 = \left(1 + \frac{2\Phi}{c^2}\right) dt^2.$$

Take the positive square root:

$$d\tau = \sqrt{1 + 2\Phi/c^2} dt.$$

Expand to first order in Φ/c^2 :

$$d\tau \approx \left(1 + \frac{\Phi}{c^2}\right) dt.$$

Take ratio of two clocks at Φ_A and Φ_B (assuming $\Phi_B > \Phi_A$):

$$\frac{d\tau_B}{d\tau_A} = \frac{1 + \Phi_B/c^2}{1 + \Phi_A/c^2} \approx 1 + \frac{\Phi_B - \Phi_A}{c^2}.$$

Clocks at higher gravitational potential run **faster**. By the equivalence principle, a uniformly accelerated frame is locally indistinguishable from a uniform Newtonian field $\Phi = gz'$. Substitute $\Phi_h - \Phi_0 = gh$:

$$\frac{d\tau_h}{d\tau_0} = 1 + \frac{gh}{c^2},$$

recovering (6). The two effects are the **same physical phenomenon**: the rate of an ideal clock depends on the gravitational potential at its location, equivalently on the position in an accelerated frame.

(e) Invariant 4-acceleration

In curved coordinates $a^\mu \equiv D^2 x^\mu / D\tau^2$ denotes the absolute derivative $U^\nu \nabla_\nu U^\mu$. A static observer at fixed z' has 4-velocity with only $U^{t'}$ non-zero. Apply normalisation $g_{\mu\nu} U^\mu U^\nu = -c^2$:

$$-\left(1 + \frac{gz'}{c^2}\right)^2 c^2 (U^{t'})^2 = -c^2.$$

Solve for $U^{t'}$ (positive root):

$$U^{t'} = \frac{1}{1 + gz'/c^2}.$$

Define $\alpha \equiv 1 + gz'/c^2$, so $U^\mu = (\alpha^{-1}, 0, 0, 0)$.

Write the metric components $g_{t't'} = -\alpha^2 c^2$, $g_{z'z'} = 1$:

$$g^{z'z'} = 1, \quad g^{t't'} = -\frac{1}{\alpha^2 c^2}.$$

Differentiate $g_{t't'}$ with respect to z' :

$$\partial_{z'} g_{t't'} = -2\alpha c^2 \cdot \frac{\partial \alpha}{\partial z'}.$$

Substitute $\partial \alpha / \partial z' = g/c^2$:

$$\partial_{z'} g_{t't'} = -2\alpha c^2 \cdot \frac{g}{c^2} = -2\alpha g.$$

Compute $\Gamma_{t't'}^{z'}$ from the Christoffel formula $\Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\rho} (\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu})$:

$$\Gamma_{t't'}^{z'} = -\frac{1}{2} g^{z'z'} \partial_{z'} g_{t't'}.$$

Substitute $g^{z'z'} = 1$ and the previous result:

$$\Gamma_{t't'}^{z'} = -\frac{1}{2} (1) (-2\alpha g) = \alpha g.$$

Expand the absolute derivative:

$$a^\mu = U^\nu \partial_\nu U^\mu + \Gamma_{\nu\lambda}^\mu U^\nu U^\lambda.$$

Static observer $\Rightarrow U^\mu$ depends only on z' and $\partial_{t'} U^\mu = 0$; only $\Gamma_{t't'}^\mu (U^{t'})^2$ survives:

$$a^{z'} = \Gamma_{t't'}^{z'} (U^{t'})^2.$$

Substitute the Christoffel and 4-velocity:

$$a^{z'} = \alpha g \cdot \alpha^{-2} = \frac{g}{\alpha}, \quad a^{t'} = 0.$$

Compute the invariant magnitude squared:

$$a^2 = g_{\mu\nu} a^\mu a^\nu = g_{z'z'} (a^{z'})^2.$$

Substitute $g_{z'z'} = 1$:

$$a^2 = \frac{g^2}{\alpha^2}.$$

Take the positive square root:

$$a(z') = \frac{g}{1 + gz'/c^2}.$$

At $z' = 0$: $a_0 = g$. At $z' = h$: $a_h = g/(1 + gh/c^2) < g$. The proper acceleration to remain static **decreases with height**: the Rindler family is rigid (Born) but not of constant proper acceleration. Equivalently, by the equivalence principle, the local "gravitational" acceleration is redshifted by the same factor $1 + gh/c^2$ as the clock rate in (c).

Question 3 — Schwarzschild circular orbit and binary GW emission

Problem. (a) $d\phi/d\tau$ at the spaceship for circular Schwarzschild orbit at $R_o = 11GM_{\text{bh}}/c^2$. (b) Period at infinity. (c) Period on the spaceship. (d) Show $R_s = (T^2 c^6 / (32\pi^2 G^2 M_{\text{bh}}^2))^{1/3}$ for a binary. (e) Frequency of emitted gravitational wave.

(a) Angular velocity at the spaceship

Equatorial circular orbit: $\theta = \pi/2$, $dr/d\tau = d^2r/d\tau^2 = 0$. The radial geodesic equation reduces to

$$\Gamma_{tt}^r \left(\frac{dt}{d\tau}\right)^2 + \Gamma_{\phi\phi}^r \left(\frac{d\phi}{d\tau}\right)^2 = 0.$$

Compute the relevant Christoffels (at $\theta = \pi/2$):

$$\Gamma_{tt}^r = \frac{GM_{\text{bh}}}{r^2} \left(1 - \frac{2GM_{\text{bh}}}{c^2 r}\right), \quad \Gamma_{\phi\phi}^r = -r \left(1 - \frac{2GM_{\text{bh}}}{c^2 r}\right).$$

Substitute into the geodesic and cancel the common factor $(1 - 2GM_{\text{bh}}/c^2 r)$:

$$\frac{GM_{\text{bh}}}{r^2} \left(\frac{dt}{d\tau}\right)^2 = r \left(\frac{d\phi}{d\tau}\right)^2.$$

Divide both sides by $(dt/d\tau)^2$ and use $d\phi/dt = (d\phi/d\tau)/(dt/d\tau)$:

$$\frac{GM_{\text{bh}}}{r^2} = r \left(\frac{d\phi}{dt}\right)^2.$$

Solve for $(d\phi/dt)^2$:

$$\left(\frac{d\phi}{dt}\right)^2 = \frac{GM_{\text{bh}}}{r^3}.$$

Substitute $r = R_o = 11GM_{\text{bh}}/c^2$:

$$\left(\frac{d\phi}{dt}\right)^2 = \frac{GM_{\text{bh}} c^6}{(11GM_{\text{bh}})^3}.$$

Simplify:

$$\left(\frac{d\phi}{dt}\right)^2 = \frac{c^6}{11^3 G^2 M_{\text{bh}}^2}.$$

Take the positive root:

$$\frac{d\phi}{dt} = \frac{c^3}{11^{3/2} GM_{\text{bh}}}. \quad (7)$$

Use the metric on the orbit ($dr = 0$, $\theta = \pi/2$, divide $ds^2 = -c^2 d\tau^2$ by $d\tau^2$):

$$-c^2 = -\left(1 - \frac{2GM_{\text{bh}}}{c^2 r}\right) c^2 \left(\frac{dt}{d\tau}\right)^2 + r^2 \left(\frac{d\phi}{d\tau}\right)^2.$$

Rewrite $r^2(d\phi/d\tau)^2 = r^2(d\phi/dt)^2(dt/d\tau)^2$ using $(d\phi/dt)^2 = GM_{\text{bh}}/r^3$:

$$r^2 \left(\frac{d\phi}{d\tau}\right)^2 = \frac{GM_{\text{bh}}}{r} \left(\frac{dt}{d\tau}\right)^2.$$

Insert $r = 11GM_{\text{bh}}/c^2$, giving $GM_{\text{bh}}/r = c^2/11$ and $1 - 2GM_{\text{bh}}/(c^2 r) = 9/11$:

$$-c^2 = -\frac{9}{11} c^2 \left(\frac{dt}{d\tau}\right)^2 + \frac{c^2}{11} \left(\frac{dt}{d\tau}\right)^2.$$

Combine on the right:

$$-c^2 = -\frac{8}{11}c^2\left(\frac{dt}{d\tau}\right)^2.$$

Solve for $(dt/d\tau)^2$:

$$\left(\frac{dt}{d\tau}\right)^2 = \frac{11}{8}.$$

Multiply (7) by $dt/d\tau = \sqrt{11/8}$:

$$\frac{d\phi}{d\tau} = \frac{c^3}{11^{3/2} GM_{\text{bh}}} \sqrt{\frac{11}{8}}.$$

Combine the powers of 11 in the denominator: $11^{3/2}/\sqrt{11/8} = 11^{3/2}\sqrt{8/11}$.

$$11^{3/2}\sqrt{8/11} = \sqrt{11^3} \cdot \sqrt{8/11} = \sqrt{11^3 \cdot 8/11} = \sqrt{8 \cdot 11^2} = 11\sqrt{8} = 22\sqrt{2}.$$

Therefore:

$$\boxed{\frac{d\phi}{d\tau} = \frac{c^3}{22\sqrt{2} GM_{\text{bh}}}.} \quad (8)$$

(b) Period at infinity

Coordinate time t is the proper time of an observer at infinity. From (7):

$$\boxed{T_{\infty} = \frac{2\pi}{d\phi/dt} = 2\pi \cdot 11^{3/2} \frac{GM_{\text{bh}}}{c^3}.} \quad (9)$$

(c) Period on the spaceship

Apply (8):

$$T_{\text{ship}} = \frac{2\pi}{d\phi/d\tau} = \frac{2\pi \cdot 22\sqrt{2} GM_{\text{bh}}}{c^3}.$$

Combine the numerical constants:

$$T_{\text{ship}} = 44\sqrt{2} \pi \frac{GM_{\text{bh}}}{c^3}.$$

Cross-check using the time-dilation factor $T_{\text{ship}}/T_{\infty} = (d\tau/dt) = \sqrt{8/11}$:

$$T_{\infty}\sqrt{8/11} = 2\pi \cdot 11^{3/2} \sqrt{8/11} \frac{GM_{\text{bh}}}{c^3} = 22\sqrt{2} \cdot 2\pi \frac{GM_{\text{bh}}}{c^3} = 44\sqrt{2} \pi \frac{GM_{\text{bh}}}{c^3}.$$

Final answer:

$$\boxed{T_{\text{ship}} = 44\sqrt{2} \pi \frac{GM_{\text{bh}}}{c^3} = T_{\infty}\sqrt{\frac{8}{11}}.}$$

The orbiting clock runs slow relative to infinity by the standard Schwarzschild factor for circular orbits.

(d) Binary separation in Schwarzschild radii

Two equal masses M_{bh} separated by distance D . Adopt the Kepler form for a test particle in the field of M_{bh} at distance D (this convention matches the printed answer; physically each mass orbits the common centre at radius $D/2$, but the printed formula uses the test-particle scaling):

$$\Omega^2 = \frac{GM_{\text{bh}}}{D^3}.$$

Use $T = 2\pi/\Omega$:

$$T^2 = \frac{4\pi^2}{\Omega^2}.$$

Substitute Ω^2 :

$$T^2 = \frac{4\pi^2 D^3}{GM_{\text{bh}}}.$$

Define $R_s \equiv D/R_{\text{Sch}}$ with $R_{\text{Sch}} = 2GM_{\text{bh}}/c^2$:

$$D = R_s \frac{2GM_{\text{bh}}}{c^2}.$$

Cube:

$$D^3 = R_s^3 \frac{8G^3 M_{\text{bh}}^3}{c^6}.$$

Substitute into T^2 :

$$T^2 = \frac{4\pi^2}{GM_{\text{bh}}} \cdot R_s^3 \frac{8G^3 M_{\text{bh}}^3}{c^6}.$$

Collect constants:

$$T^2 = \frac{32\pi^2 G^2 M_{\text{bh}}^2}{c^6} R_s^3.$$

Solve for R_s^3 :

$$R_s^3 = \frac{T^2 c^6}{32\pi^2 G^2 M_{\text{bh}}^2}.$$

Take the cube root:

$$\boxed{R_s = \left(\frac{T^2 c^6}{32\pi^2 G^2 M_{\text{bh}}^2} \right)^{1/3}}. \quad (10)$$

(e) Gravitational-wave frequency

Place the orbit in the xy -plane with the centre of mass at the origin. Each black hole sits at $\pm \mathbf{r}(t)$ with $|\mathbf{r}| = D/2$:

$$\mathbf{x}_1 = \frac{D}{2}(\cos \Omega t, \sin \Omega t, 0), \quad \mathbf{x}_2 = -\mathbf{x}_1.$$

Insert into the quadrupole definition $I^{ij} = M_{\text{bh}} c^2 (x_1^i x_1^j + x_2^i x_2^j)$:

$$I^{ij} = M_{\text{bh}} c^2 (x_1^i x_1^j + (-x_1^i)(-x_1^j)) = 2M_{\text{bh}} c^2 x_1^i x_1^j.$$

Compute the xx component:

$$I^{xx} = 2M_{\text{bh}} c^2 \cdot \frac{D^2}{4} \cos^2(\Omega t).$$

Simplify:

$$I^{xx} = \frac{M_{\text{bh}} c^2 D^2}{2} \cos^2(\Omega t).$$

Apply $\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$:

$$I^{xx} = \frac{M_{\text{bh}} c^2 D^2}{4} [1 + \cos(2\Omega t)].$$

Differentiate once in t :

$$\dot{I}^{xx} = -\frac{M_{\text{bh}} c^2 D^2}{4} \cdot 2\Omega \sin(2\Omega t).$$

Differentiate again:

$$\ddot{I}^{xx} = -\frac{M_{\text{bh}} c^2 D^2}{4} \cdot 4\Omega^2 \cos(2\Omega t) = -M_{\text{bh}} c^2 D^2 \Omega^2 \cos(2\Omega t).$$

The radiated metric perturbation $\bar{h}^{ij} \propto \ddot{I}^{ij}$ oscillates at 2Ω . Read off the frequency:

$$f_{\text{GW}} = \frac{2\Omega}{2\pi} = \frac{\Omega}{\pi} = \frac{2}{T}. \quad (11)$$

Plug in $T = 5 \times 10^4 \text{ s}$:

$$f_{\text{GW}} = \frac{2}{5 \times 10^4 \text{ s}}.$$

Evaluate:

$$f_{\text{GW}} = 4 \times 10^{-5} \text{ Hz}.$$

This sits in the LISA band, characteristic of supermassive-binary inspirals.

Question 4 — Stress-energy tensor of an ideal fluid

Problem. (a) Why must the source $T^{\mu\nu}$ be a rank-2 tensor (Lorentz boost of dust). (b) Construct $T^{\mu\nu}$ from ρ and U^μ and write $T^{\mu\mu}$. (c) Interpret $i = j$ entries as pressure and explain off-diagonal vanishing. (d) Show $T^{\mu\nu} = (\rho + P/c^2)U^\mu U^\nu + P\eta^{\mu\nu}$ and uniqueness. (e) Replace $\eta^{\mu\nu} \rightarrow g^{\mu\nu}$ in curved space. (f) Show $A^\mu = U^\nu U^\mu_{;\nu}$ is orthogonal to U^μ .

(a) Why a rank-2 tensor

In the rest frame S of dust at A , the matter density is the scalar $\rho = n_0 m_0$, where n_0 is the rest-frame number density. Boost to a frame S' in which the fluid moves with velocity $\mathbf{u}$, Lorentz factor γ . Number density transforms by length contraction along $\mathbf{u}$:

$$n' = \gamma n_0.$$

Rest mass is invariant, but the relativistic mass picks up γ :

$$m'_0 = m_0, \quad m'_{\text{rel}} = \gamma m_0.$$

Combine into the energy density:

$$\rho' c^2 = n' m'_{\text{rel}} c^2 = (\gamma n_0)(\gamma m_0) c^2.$$

Simplify:

$$\rho' c^2 = \gamma^2 \rho c^2.$$

Two factors of γ appear: **one each** from length contraction (volume) and from energy/mass increase. A scalar carries no boost factor; a 4-vector carries one. Two factors of γ require **two factors** of $\partial x'^\mu / \partial x^\nu$, i.e. a rank-2 (contravariant) tensor:

$$T'^{\mu\nu} = \frac{\partial x'^\mu}{\partial x^\alpha} \frac{\partial x'^\nu}{\partial x^\beta} T^{\alpha\beta}.$$

(b) Dust stress-energy tensor

Bilinear in the only available rank-1 object, the 4-velocity U^μ , and proportional to the scalar ρ :

$$\boxed{T^{\mu\nu} = \rho U^\mu U^\nu.} \quad (12)$$

Write the boosted 4-velocity:

$$U^\mu = \gamma(c, \mathbf{u}), \quad U^0 = \gamma c, \quad U^i = \gamma u^i.$$

Insert into $T^{\mu\nu} = \rho U^\mu U^\nu$:

$$T^{00} = \rho(\gamma c)^2 = \rho \gamma^2 c^2, \quad T^{ii} = \rho(\gamma u^i)^2 = \rho \gamma^2 (u^i)^2.$$

Energy density: $T^{00}/c^2 = \gamma^2 \rho$, recovering the result of (a).

(c) Pressure and off-diagonal entries

In the instantaneous rest frame, momentum-flux T^{ij} is the rate at which the i -component of momentum crosses unit area normal to $\hat{j}$. For an isotropic fluid: **(i)** no preferred direction $\Rightarrow T^{ij}$ proportional to δ^{ij} in 3D; **(ii)** T^{ii} is the normal force per unit area on a face, which is the definition of **pressure** P . Off-diagonal T^{ij} ($i \neq j$) would be a shear stress; absent for a perfect fluid (no viscosity, no momentum flux parallel to the face). $T^{0i} = 0$ because there is no bulk energy flux in the rest frame.

(d) Boost to general frame

Use signature $(-, +, +, +)$ (cover-sheet convention $-c^2 dt^2 + \dots$, so $U_\mu U^\mu = -c^2$). In the rest frame $U^\mu = (c, 0, 0, 0)$, hence the raised $U^\mu U^\nu = c^2 \text{diag}(1, 0, 0, 0)$ and $\eta^{\mu\nu} = \text{diag}(-1, 1, 1, 1)$.

Try the ansatz with two unknown scalars A, B :

$$T^{\mu\nu} = A U^\mu U^\nu + B \eta^{\mu\nu}.$$

Read off the 00 component in the rest frame:

$$T^{00} = A c^2 + B(-1) = \rho c^2.$$

Read off the ii component:

$$T^{ii} = A \cdot 0 + B \cdot 1 = P.$$

Identify B :

$$B = P.$$

Substitute into the 00 equation:

$$A c^2 - P = \rho c^2.$$

Solve for A :

$$A = \rho + \frac{P}{c^2}.$$

Substitute back into the ansatz:

$$\boxed{T^{\mu\nu} = \left(\rho + \frac{P}{c^2}\right) U^\mu U^\nu + P \eta^{\mu\nu}.} \quad (13)$$

Uniqueness. The only two rank-2 tensors that can be built from a single 4-vector U^μ and the metric are $U^\mu U^\nu$ and $\eta^{\mu\nu}$ (any higher product is excluded by rank, and $U^\mu U^\nu U^\rho U^\sigma \eta_{\rho\sigma} \propto U^\mu U^\nu$ is degenerate). Two coefficients (A, B) are fully determined by the two independent rest-frame components $(\rho c^2, P)$. Hence (13) is the unique form.

(e) Curved-space upgrade

Equivalence principle: at any event one can erect a local inertial frame in which $g^{\mu\nu} \rightarrow \eta^{\mu\nu}$ and the tensor (13) holds in its flat form. Both sides of the equation are tensors, so a tensor equation valid in one coordinate system is valid in all coordinate systems. Replacing $\eta^{\mu\nu} \rightarrow g^{\mu\nu}$ promotes the formula to the curved manifold (**minimal coupling** / comma-to-semicolon rule):

$$T^{\mu\nu} = \left(\rho + \frac{P}{c^2}\right) U^\mu U^\nu + P g^{\mu\nu}.$$

This is the unique tensorial extension reducing to (13) in the local Minkowski frame.

(f) $A^\mu \perp U^\mu$

State the normalisation of the 4-velocity:

$$U^\mu U_\mu = -c^2.$$

Differentiate covariantly along the worldline ($U^\nu \nabla_\nu$):

$$U^\nu \nabla_\nu (U^\mu U_\mu) = 0.$$

Apply the Leibniz rule:

$$U^\nu (\nabla_\nu U^\mu) U_\mu + U^\nu U^\mu (\nabla_\nu U_\mu) = 0.$$

Use metric compatibility $\nabla_\nu g_{\mu\rho} = 0$ to write $\nabla_\nu U_\mu = g_{\mu\rho} \nabla_\nu U^\rho$:

$$U^\nu U^\mu \nabla_\nu U_\mu = U^\nu U^\mu g_{\mu\rho} \nabla_\nu U^\rho = U^\nu U_\rho \nabla_\nu U^\rho.$$

Relabel $\rho \rightarrow \mu$ in this second term:

$$U^\nu U^\mu \nabla_\nu U_\mu = U^\nu U_\mu \nabla_\nu U^\mu.$$

Combine the two identical contributions:

$$2 U_\mu U^\nu \nabla_\nu U^\mu = 0.$$

Identify $A^\mu = U^\nu \nabla_\nu U^\mu = U^\nu U^\mu_{;\nu}$:

$$U_\mu A^\mu = 0.$$

The 4-acceleration is always orthogonal (Minkowski sense) to the 4-velocity. Geometrically U^μ has fixed length $-c^2$, so any change must be perpendicular.